

Final Goal-Based Document

Module 1: Public Dental Support Chatbot UI Foundation

1. Module Name

Module 1: Public Dental Support Chatbot UI Foundation

2. Main Goal

Build a **modern, premium, user-friendly, and client-attractive chatbot UI** for a public dental support website.

The chatbot UI should look highly professional and suitable for a healthcare/dental business. It should not look like a simple or basic chatbot widget. It should feel like a polished AI-powered customer support assistant that can impress the client during the first review.

At this stage, the chatbot does **not** need backend API, OpenAI, Qdrant, document upload, document chunking, or real RAG functionality. The complete goal of this module is to create a **frontend-only chatbot UI foundation** using dummy/static responses.

3. Project Context

We are building a **Public Dental Support RAG Chatbot**.

The chatbot will help public users understand general dental support topics such as:

- Dental claims
- Dental procedure codes
- Claim denials
- Claim rejections
- Claim processing issues
- Dental insurance terms
- Required claim documents
- Pre-authorization
- Tooth number requirements
- Tooth surface requirements
- Common reasons why dental claims fail
- General next steps when a claim has an issue

The chatbot must be public-safe. It must not access or process private user records.

The chatbot must not:

- Access patient records
- Check real claim status
- Ask for subscriber ID
- Ask for claim ID
- Ask for insurance ID
- Ask for Social Security number
- Ask for date of birth
- Access insurance accounts
- Access internal billing data
- Give final clinical, legal, or insurance decisions

The chatbot should only provide **general dental claims and coding guidance**.

4. Main Design Requirement

The UI must be:

- Modern
- Clean
- Professional
- Client-attractive
- Healthcare-friendly
- Premium-looking
- Smooth and interactive
- Mobile responsive
- Easy to use
- Trustworthy for public users

The design should feel like a real production-ready chatbot that can be shown to a client with confidence.

The developer has full authority to add modern UI improvements, animations, micro-interactions, hover effects, transitions, and polished visual elements as long as they improve the professional quality of the chatbot.

5. Technology Stack

Use the following frontend stack:

- Next.js
- React

- TypeScript
- Tailwind CSS
- shadcn/ui
- Lucide React icons
- Framer Motion for smooth animations and modern UI transitions

No backend integration is required in this module.

6. Working Folder

Create all chatbot UI components inside this folder:

```
components/chat/
```

Required component files:

```
ChatLauncher.tsx  
ChatWidget.tsx  
ChatHeader.tsx  
MessageList.tsx  
MessageBubble.tsx  
ChatInput.tsx  
SuggestedQuestions.tsx  
DisclaimerBanner.tsx  
SourceCitations.tsx  
HumanHandoff.tsx  
TypingIndicator.tsx  
ChatEmptyState.tsx
```

Additional helper files can be added if needed:

```
chatTypes.ts  
dummyChatData.ts  
chatUtils.ts
```

7. UI Style Direction

The chatbot should follow a premium dental/healthcare design style.

Recommended visual direction:

- White or soft off-white background
- Soft blue, teal, cyan, or dental-care accent colors
- Rounded corners
- Subtle glassmorphism effect if suitable
- Soft shadows
- Clean borders
- Smooth gradient accents
- Professional icons
- Calm and trustworthy color palette
- Clear typography
- Good spacing
- No clutter
- No harsh colors
- No childish visuals

The UI should look friendly but still professional.

8. Animation and Interaction Requirements

Use Framer Motion or similar modern animation patterns for a high-quality experience.

Add smooth animations for:

- Chat launcher hover
- Chat widget open and close
- Message appearance
- Suggested question chip hover
- Typing indicator
- Source citation reveal
- Human handoff card reveal
- Button hover and tap states

The animations should be subtle and professional. They should improve the user experience without making the UI feel heavy or distracting.

9. Required Components and Goals

9.1 ChatLauncher.tsx

Create a floating chatbot launcher button fixed at the bottom-right of the website.

Requirements

- Premium floating button
- Rounded design
- Dental/AI support icon
- Smooth hover animation
- Small pulse or glow effect if suitable
- Desktop label: "Need help with dental claims?"
- Mobile can use compact icon-only version
- Opens the chatbot widget when clicked
- Should feel modern and client-attractive

Visual Goal

The launcher should immediately look like a polished support assistant, not a default button.

9.2 ChatWidget.tsx

Create the main chatbot container.

Requirements

- Fixed bottom-right layout on desktop
- Full-screen or near full-screen layout on mobile
- Card-style premium design
- Smooth open/close animation
- Contains:
 - ChatHeader
 - DisclaimerBanner
 - MessageList
 - SuggestedQuestions
 - ChatInput
- Proper height management
- Professional visual hierarchy
- Clean spacing
- Responsive behavior
- No overflow issues

Desktop Behavior

The widget should appear as a premium floating chat card.

Mobile Behavior

The widget should become full-screen or nearly full-screen for better usability.

9.3 ChatHeader.tsx

Create the chatbot header.

Content

```
Dental Support Assistant  
General dental claims guidance
```

Requirements

- Assistant avatar or icon
- Status indicator such as “Online”
- Close/minimize button
- Premium healthcare-style header
- Clear branding
- Soft gradient or subtle background if suitable
- Professional layout

Visual Goal

The header should clearly establish trust and assistant identity.

9.4 DisclaimerBanner.tsx

Create a small public-safety disclaimer banner.

Text

```
This assistant provides general dental claims and coding guidance only. It  
cannot check personal claim status or access private insurance records.
```

Requirements

- Small info icon
 - Soft background
 - Professional tone
 - Easy to read
 - Should not feel alarming
 - Should not dominate the interface
 - Should be visible near the top of the widget
-

9.5 SuggestedQuestions.tsx

Create clickable suggested question chips.

Suggested Questions

```
Why was my dental claim denied?  
What is a dental procedure code?  
What does pre-authorization mean?  
What is the difference between rejected and denied?  
Why is tooth number required?  
Why is tooth surface required?  
What documents may be needed for a dental claim?  
Why did insurance not pay the full amount?
```

Requirements

- Rounded pill buttons
 - Clean hover effects
 - Smooth animations
 - Wrap properly on small screens
 - Clicking a question sends it as a user message
 - Suggested questions should help guide public users
 - Should look clean and modern
-

9.6 MessageList.tsx

Create the scrollable message area.

Requirements

- Shows welcome message when conversation starts
 - Shows user and assistant messages
 - Auto-scrolls to latest message
 - Supports loading/typing state
 - Clean spacing between messages
 - Mobile-friendly scrolling
 - No layout jump issues
 - Should support dummy messages for now
-

9.7 MessageBubble.tsx

Create a reusable message bubble component.

Requirements

- User messages aligned right
- Assistant messages aligned left
- Different visual design for user and assistant
- Assistant messages should support multi-line text
- Assistant messages should support bullet-style formatting if needed
- Smooth message entry animation
- Clean readable text
- Professional bubble styling

Visual Direction

User bubble can use the main accent color. Assistant bubble should use a soft neutral background with subtle border and professional spacing.

9.8 ChatInput.tsx

Create the input area.

Requirements

- Textarea input
- Send button
- Enter to send
- Shift + Enter for new line
- Send button disabled when input is empty
- Input disabled while dummy response is loading
- Clean focus state
- Smooth button interaction
- Mobile-friendly size

Placeholder Text

Ask about dental claims, codes, denials, or insurance terms...

9.9 SourceCitations.tsx

Create a source citation display component.

For Module 1, use dummy sources.

Dummy Sources

Dental Claims FAQ
Common Claim Failure Reasons
Dental Insurance Glossary

Requirements

- Show below assistant response when sources are available
- Use small badges or compact source cards
- Should look professional and non-intrusive
- Should improve trust in the answer
- Should be easy to replace later with real RAG sources

9.10 HumanHandoff.tsx

Create a support handoff card.

Text

For exact claim status or personal insurance details, please contact your dental office or insurance carrier.

Button

Contact Support

Requirements

- Display when the dummy logic detects a private claim-status request
- Clean card design
- Professional call-to-action
- Friendly but firm message
- Should guide users safely without asking for private details

9.11 TypingIndicator.tsx

Create a premium typing/loading indicator.

Requirements

- Show when assistant is preparing dummy response
 - Smooth animated dots
 - Assistant-style bubble
 - Should feel modern and polished
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9.12 ChatEmptyState.tsx

Create a polished empty/welcome state.

Requirements

- Friendly welcome message
- Small assistant icon
- Short introduction
- Suggested questions visible below
- Clean layout

Welcome Message

Hi, I'm your Dental Support Assistant. I can help explain general dental claims, coding, insurance terms, claim denials, and common processing issues. I can't access personal claim records, but I can guide you on what information may be needed and what steps to take next.

10. Dummy Chat Behavior

For this module, use frontend-only dummy responses.

When the user sends a message:

1. User message appears immediately
2. Typing indicator appears
3. After a short delay, assistant dummy response appears
4. Dummy sources appear under the assistant response

General Dummy Response

A dental claim may be denied for several general reasons, such as missing information, inactive coverage, waiting periods, frequency limits, or missing

required documentation. Please review your Explanation of Benefits or contact your dental office or insurance carrier for exact claim details.

Dummy Sources

Common Claim Failure Reasons
Dental Claims FAQ

11. Safe Handoff Behavior

If the user asks for:

- Claim status
- Claim ID check
- Subscriber ID help
- Insurance ID lookup
- Patient record access
- Exact private claim reason
- Personal insurance details

Then show this safe response:

I'm sorry, but I can't check personal claim status or access private insurance records. For exact claim details, please contact your dental office, billing team, or insurance carrier.

Also display the HumanHandoff component.

12. Private Request Detection Keywords

For dummy frontend logic, detect keywords such as:

claim status
check my claim
claim id
subscriber id
insurance id
policy number
patient record
my record
date of birth

ssn
social security
personal insurance

If any of these are detected, return the safe handoff response.

13. Mobile Responsiveness Requirements

The chatbot must work perfectly on mobile.

Mobile Requirements

- Widget should be full-screen or near full-screen
 - Input should remain at the bottom
 - Message list should scroll smoothly
 - Suggested question chips should wrap properly
 - Buttons should be easy to tap
 - Text should remain readable
 - No horizontal scroll
 - No hidden input field
 - Header should remain clean
 - Close button should be easily accessible
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14. Accessibility Requirements

The UI should be accessible and user-friendly.

Requirements:

- Buttons should have clear labels
 - Good color contrast
 - Keyboard-friendly input
 - Focus states should be visible
 - Icons should not be the only source of meaning
 - Text should be readable on mobile and desktop
 - Form should not submit empty messages
-

15. Client-Attractive Enhancements

The developer has permission to add professional enhancements such as:

- Soft gradient header

- Animated assistant avatar
- Pulse effect on launcher
- Floating badge such as “AI Support”
- Smooth message animations
- Premium loading indicator
- Source badges
- Support handoff card
- Subtle shadows and borders
- Clean empty state
- Modern micro-interactions
- Polished mobile drawer experience

The final UI should feel impressive during a client demo.

16. Success Criteria

Module 1 is complete when:

- Chat launcher appears on the website
 - Chat launcher looks modern and premium
 - Chat widget opens and closes smoothly
 - Chat UI looks professional and client-ready
 - Welcome message appears
 - Suggested questions are visible and clickable
 - User can type and send a message
 - User message appears immediately
 - Typing indicator appears before assistant response
 - Dummy assistant response appears
 - Source citations appear under assistant response
 - Disclaimer banner is visible
 - Safe handoff message appears for private claim-status questions
 - Human handoff card appears when needed
 - UI works smoothly on desktop
 - UI works smoothly on mobile
 - Code is component-based
 - Code is clean and reusable
 - No backend dependency exists in this module
 - No OpenAI or Qdrant connection exists in this module
 - Output is ready for client visual review
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17. Important Development Boundaries

Do not build backend API in this module.

Do not connect OpenAI in this module.

Do not connect Qdrant in this module.

Do not build document upload in this module.

Do not add real RAG response in this module.

Do not collect private user information.

Do not ask users for claim IDs, subscriber IDs, insurance IDs, or patient information.

This module is only for building the professional frontend chatbot UI foundation with dummy responses.

18. Final Delivery Expectation

The final output of Module 1 should be a **highly polished, modern, frontend-only chatbot UI** that can be shown to a client as the first visual demo.

The UI should make the client feel that the chatbot is:

- Professional
- Trustworthy
- Modern
- Easy to use
- Healthcare suitable
- Ready for future AI/RAG integration

The first visual impression is very important. The chatbot should look like a premium customer support assistant, not a basic test widget.